

Project Title: Computer Pricing: An Exploratory Analysis

Scenario:

You are a data analyst for a computer magazine in the mid-1990s. Your editor has given you a dataset (**Computer_Data.csv**) of new personal computers and their configurations. She wants you to write a short report that helps readers understand the current market.

Your Task:

Using your R skills, analyze the dataset and answer the following questions in a short report (e.g., 3-4 pages with plots). Your audience is the magazine's readers, who are tech-savvy but not necessarily statisticians.

Guiding Questions:

1. Data Summary and Structure:

- What are the variables in the dataset? Classify each as numerical or categorical.
- Provide a summary (mean, median, min, max) for the key numerical variables: price, speed, hd, ram, screen, ads.

2. Univariate Analysis (Single Variables):

- Create a histogram for price. What does the distribution of computer prices look like? Is it symmetric or skewed?
- Create histograms for speed, hd, and ram. How are these technical specifications distributed?

3. Bivariate Analysis (Relationships between Two Variables):

- Is there a relationship between price and speed? Create a scatter plot and calculate the correlation and covariance. What does this tell you?
- Repeat the analysis for price vs. hd and price vs. ram. Which specification seems most strongly correlated with price?
- Compare the price of computers with a CD-ROM (cd=yes) versus those without (cd=no). What do you observe?

4. Initial Insights and Conclusion:

- Based on your analysis, what was a typical price for a computer in this dataset?
- If a reader wanted a fast computer with a large hard drive, what could they expect to pay?
- What was the most important factor influencing the price of a computer? Justify your answer with the evidence from your analysis.